

TECHNICAL SPECIFICATIONS

SECTION 21 22 00

CLEAN AGENT FIRE SUPPRESSION

Project: Meridian Data Centre Campus, Phase 1
Navi Mumbai, India
Revision: 2024 (December 12, 2024)

PART 1 - GENERAL

1.1 Summary

This section specifies engineered gaseous clean agent fire suppression systems, detection controls, and release valves for server rooms.

1.2 References

- NFPA 2001 - Standard on Clean Agent Fire Extinguishing Systems
- UL 2166 - Standard for Halocarbon Clean Agent Extinguishing System Units
- IS 15493 - Gaseous fire extinguishing systems

1.3 Submittal Requirements

A. Contractor shall submit the following for Engineer's approval prior to equipment procurement:

1. Product Data: Technical datasheets detailing all electrical and physical parameters, performance curves, and compliance.
2. Shop Drawings: Dimensional layouts showing clearance requirements, cabling paths, and weights.
3. Compliance Matrix: Clause-by-clause confirmation of conformance. Any deviation must be explicitly noted.
4. Factory Test Certificates and Type Test Certifications.

1.4 Quality Assurance

A. Equipment shall be manufactured by a firm specializing in the equipment class with a minimum of twenty (25) years of continuous commercial manufacturing history.

B. As specified in Section 26 05 00 Part 1.4.3.B, all medium voltage switchgear and high-power electrical components must be supplied with certified test certificates from an internationally recognized independent laboratory (ASTA, KEMA, or equivalent) confirming compliance with IEC standards.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers

A. Subject to compliance with technical requirements, approved manufacturers are limited to the following:

2.2 Performance and Design Requirements

agent: Gaseous suppressant agent must be FK-5-1-12 (3M Novec 1230).

discharge_time: Agent discharge must complete within 10.0 seconds or less.

hold_time: Agent design concentration must be retained in the protected hazard zone for ≥ 10 minutes.

safety: Design concentration must maintain an acceptable safety margin below the NOAEL limit (10.0%).

Injected Contract Defects (Internal Quality Control Audit Targets):

- **DEF-006:** Clean Agent Fire Suppression spec Section 21 22 00 Part 1.2.1 references 'NFPA 2001 (2008 Edition)', which is a withdrawn standard revision. (Location: 21 22 00 Part 1.2.1)

PART 3 - EXECUTION

3.1 Installation and Examination

- A. Verify that structural concrete floor slab is fully cured. Verify structural load limit of 2,800 kg/m² is not exceeded in UPS or battery rooms.
- B. Maintain a minimum clear work clearance of 1000 mm in front of all enclosures, and 1200 mm above, in accordance with local Indian Electricity Rules and IEC standards.

3.2 Field Quality Control and Site Commissioning

- A. Startup services shall be performed by factory-authorized technicians.
- B. Commissioning tests must be scheduled and carried out in accordance with the project's overall Cx Test Register. Every testing step must map back to a specific specification clause.

door_fan_test: Perform room integrity door fan testing to verify agent retention hold time.

discharge_test: Conduct puff testing with nitrogen to verify piping nozzle flow.